

Replica Bounds for the PDE Encoder and Linear-Attention Decoder

1 Claim

The exact task-level correspondence with the low-rank ICL regression framework of [1] is:

$$\text{one task} \iff \text{one coefficient vector } z,$$

and one prompt for that task is the whole collection

$$\mathcal{C}_z = \{(f_i, u_i)\}_{i=1}^m.$$

The weak rows inside the prompt are examples for the same task; they are not independent tasks. There are two uses of the same replica framework.

1. **Encoder ICL.** The weak-form encoder is trained so that the produced system is consistent:

$$\widehat{G}z - \widehat{b} \approx 0.$$

This is a low-rank regression generalization error at the level of a full prompt for a fixed z .

2. **Decoder ICL.** The decoder is taken to be *linear* attention, not softmax attention. Linear attention implements

$$\widehat{z} = W_{\text{dec}} \left(\frac{K}{M} \widehat{G}^\top \widehat{b} \right), \quad M = mR,$$

so W_{dec} is a learned spectral preconditioner for the empirical Gram matrix. The paper's replica solution is precisely a distributional formula for this learned preconditioner.

The resulting end-to-end bound has the exact form

$$\boxed{\frac{1}{K} \mathbb{E} \|\widehat{z} - z\|_2^2 \lesssim E_{\text{dec}}^{\text{rep,GP}} + B_{\text{dec}}^2 C_{\text{GP}} E_{\text{enc}}^{\text{rep,GP}}}$$

and

$$\boxed{\mathbb{E} \frac{\|\widehat{u}_\star - u_\star\|_2^2}{\|u_\star\|_2^2} \lesssim C_{\text{elliptic}}^2 B_\Psi^2 K \left(E_{\text{dec}}^{\text{rep,GP}} + B_{\text{dec}}^2 C_{\text{GP}} E_{\text{enc}}^{\text{rep,GP}} \right) + \text{dictionary/gauge bias.}}$$

Here $E_{\text{enc}}^{\text{rep,GP}}$ and $E_{\text{dec}}^{\text{rep,GP}}$ denote the replica errors for the exact GP-induced prompt law. They reduce to the closed forms in the paper, with D replaced by K , only in the special case where the weak-row design is task-independent and isotropic.

2 Exact Gaussian PDE Prompt Model

We work with the discretized elliptic family

$$A(z) = A_0 + \sum_{k=1}^K z_k A_k, \quad A(z)u = f.$$

Assume a Gaussian low-rank latent law

$$z = \sqrt{\frac{K}{r_z}} B\eta, \quad \eta \sim N(0, I_{r_z}), \quad B^\top B = I_{r_z},$$

so the task rank is r_z and $\rho_z = r_z/K$. The full-rank Gaussian case is included by taking $r_z = K$.

Assume the forcing is a Gaussian process. After discretization,

$$f_i \sim N(0, \Sigma_f), \quad u_i = A(z)^{-1} f_i.$$

For a test function v_r , define the exact weak row and label, with $j = (i, r)$,

$$g_{j,k} := \langle A_k u_i, v_r \rangle, \quad b_j := \langle f_i - A_0 u_i, v_r \rangle.$$

Then, by the PDE,

$$b_j = g_j^\top z.$$

Thus, for a fixed task z , the whole prompt induces the exact linear system

$$G_z z = b_z, \quad G_z \in \mathbb{R}^{M \times K}, \quad M = mR.$$

Conditionally on z , every row g_j is Gaussian because it is a linear image of the Gaussian forcing f_i :

$$g_j \mid z \sim N(0, H_r(z)), \quad H_r(z) = B_r A(z)^{-1} \Sigma_f A(z)^{-\top} B_r^\top,$$

where $(B_r u)_k = \langle A_k u, v_r \rangle$. The exact task-conditional prompt law is therefore

$$P_z(dG, db) = P_z(dG) \delta_{Gz}(db),$$

where $P_z(dG)$ is the Gaussian law determined by the matrices $\{H_r(z)\}_{r=1}^R$ and the repeated forcing draws.

Proposition 1 (exact correspondence). *The PDE ICL training distribution is exactly a low-rank task distribution over prompts:*

$$z^\mu \sim N(0, \Sigma_z), \quad C^\mu \sim P_{z^\mu}, \quad b^\mu = G^\mu z^\mu.$$

The task is z^μ ; the prompt is the whole matrix-vector pair (G^μ, b^μ) generated from the many PDE pairs $\{(f_i^\mu, u_i^\mu)\}_{i=1}^m$.

Proof. For every i, r ,

$$b_{(i,r)} = \langle f_i - A_0 u_i, v_r \rangle = \left\langle \sum_{k=1}^K z_k A_k u_i, v_r \right\rangle = \sum_{k=1}^K z_k g_{(i,r),k}.$$

Stacking all mR rows gives $b = Gz$. Since f_i is Gaussian and $u_i = A(z)^{-1} f_i$, G is conditionally Gaussian given z , with covariance given above. This proves the prompt-level correspondence. $\square$

Remark 1. *With $f \sim GP$, the design law $P_z(dG)$ depends on the task through $A(z)^{-1}$. Therefore the exact GP-forcing model is not the paper’s task-independent iid Gaussian design unless $H_r(z)$ is independent of z and isotropic after normalization. The paper is recovered exactly in that special case, or in the alternative data-generation convention where u_i is sampled from a task-independent GP and $f_i = A(z)u_i$.*

3 Task Diversity and GP-Induced Difficulty

The exact task diversity object is the empirical covariance of the distinct pretraining tasks:

$$S_z := \frac{1}{M_0} \sum_{\mu=1}^{M_0} z^\mu (z^\mu)^\top.$$

In the population model, S_z concentrates around $\Sigma_z = \mathbb{E}[zz^\top]$. Its rank, or more precisely its spectrum, is the exact analogue of the paper’s low-rank task matrix. Thus

$$r_z = \text{rank}(\Sigma_z), \quad \rho_z = \frac{r_z}{K}, \quad \kappa_z = \frac{M_0}{K}.$$

The task-diversity transition is

$$M_0 \simeq r_z \quad \text{or equivalently} \quad \kappa_z \simeq \rho_z.$$

The GP forcing does not create new tasks. It determines how well each task is observed through the prompt. The exact observability, or prompt difficulty, is the collection of conditional row covariances

$$\mathcal{H} := \{H_r(z) : z \in \text{supp}(\Sigma_z), r = 1, \dots, R\}, \quad H_r(z) = B_r A(z)^{-1} \Sigma_f A(z)^{-\top} B_r^\top.$$

For a scalar summary one may inspect the spectrum of

$$H(z) := \frac{1}{R} \sum_{r=1}^R H_r(z),$$

especially restricted to the active task subspace $\text{range}(\Sigma_z)$. Small eigenvalues of this restricted matrix mean that some coefficient directions are almost invisible to the GP prompts. This is difficulty, not diversity.

4 Replica Error Reused from the Paper

Let $\Pi \in \{\text{TM}, \text{IDG}, \text{ODG}\}$ denote task memorization, in-distribution generalization, or out-of-distribution generalization. In the notation of the paper, the exact task correspondence is

$$D \mapsto K, \quad r \mapsto r_z, \quad \rho = r/D \mapsto \rho_z = r_z/K, \quad \kappa = M_0/D \mapsto \kappa_z = M_0/K.$$

For the exact GP prompt law $P_z(dG, db)$, the same replica construction applies to a Hamiltonian averaged over the joint law (z, G, b) with $b = Gz$. Denote the resulting scalar errors by $E_{\text{TM}}^{\text{rep,GP}}$, $E_{\text{IDG}}^{\text{rep,GP}}$, and $E_{\text{ODG}}^{\text{rep,GP}}$.

If, in addition, the prompt design is task-independent and isotropic, $H_r(z) = I_K/K$, the exact GP errors reduce to the paper’s closed-form low-rank-design errors

$$E_{\text{TM}}^{\text{rep}} = 1 - 2m + q + \frac{\bar{q}}{\alpha}, \quad E_{\text{IDG}}^{\text{rep}} = 1 - 2m_0 + q_0 + \frac{\bar{q}}{\alpha}, \quad E_{\text{ODG}}^{\text{rep}} = 1 - 2\bar{m} + \bar{q} + \frac{\bar{q}}{\alpha},$$

where $(q, m, q_0, m_0, \bar{q}, \bar{m})$ solve the same self-consistent equations as in Appendix E of [1]. In the noiseless large-pretraining limit $\alpha \rightarrow \infty$, $\sigma = 0$, $\kappa_z \gamma > 1$, the paper’s closed form becomes

$$E_{\text{TM}}^{\text{rep}} = \frac{\min(\rho_z, \kappa_z)}{\tilde{\alpha}} \left(1 + \frac{1 - \min(\rho_z, \kappa_z)}{\kappa_z \gamma - 1} \right),$$

and

$$E_{\text{IDG}}^{\text{rep}} = \begin{cases} E_{\text{TM}}^{\text{rep}}, & \rho_z < \kappa_z, \\ 1 - \frac{\kappa_z}{\rho_z} - \frac{\kappa_z(\kappa_z - \rho_z)}{\rho_z(\kappa_z \gamma - 1)} + E_{\text{TM}}^{\text{rep}}, & \rho_z \geq \kappa_z. \end{cases}$$

This is the phase transition: below $\rho_z = \kappa_z$ the learned map exploits the low-rank task structure; above it, performance is bottlenecked by task diversity.

5 Encoder as Low-Rank ICL

The encoder is trained from PDE prompts to output weak rows and labels $(\widehat{G}, \widehat{b})$. Its relevant generalization metric is not coefficient error but weak consistency:

$$\mathcal{E}_{\text{enc}} := \frac{1}{M} \mathbb{E} \|\widehat{G}z - \widehat{b}\|_2^2.$$

For each fixed task z , all $M = mR$ rows in the prompt obey $b_j = g_j^\top z$. If the encoder is implemented by a one-layer linear attention model, its scalar prediction for a queried weak row q has the form

$$\widehat{b}(q) = \text{tr}(W_{\text{enc}} H_q^\top), \quad H_q = \frac{K}{L_{\text{enc}}} q \sum_{\ell \leq L_{\text{enc}}} b_\ell x_\ell^\top,$$

with x_ℓ the representation of the ℓ -th weak row. The task variable used by the replica calculation is z ; the full prompt generated from $P_z(dG, db)$ is the context for that task.

Proposition 2 (encoder residual bound). *Suppose the encoder uses linear attention and is pre-trained on the exact GP-induced prompt law of Proposition 1. Under the replica saddle-point prediction for this exact law,*

$$\mathcal{E}_{\text{enc}} = \frac{1}{M} \mathbb{E} \|\widehat{G}z - \widehat{b}\|_2^2 \leq E_{\Pi, \text{enc}}^{\text{rep, GP}}.$$

When $H_r(z) = I_K/K$ exactly, this is the paper's $E_{\Pi, \text{enc}}^{\text{rep}}$ with $D = K$ and $r = r_z$.

Proof. By Proposition 1, each prompt is sampled by first drawing a single task z , then drawing the whole matrix-vector pair $(G, b) \sim P_z$ with $b = Gz$. Thus every scalar residual component is a prediction error for the same task z :

$$\widehat{b}_j - \widehat{g}_j^\top z.$$

The replicated partition function for the encoder is therefore the paper's linear-attention partition function with $D = K$, but with the example distribution inside each context replaced by the exact conditional law $P_z(dG, db)$. The saddle-point prediction of this replicated system is, by definition, $E_{\Pi, \text{enc}}^{\text{rep, GP}}$. Averaging the M scalar residuals inside a prompt gives the displayed bound. If $H_r(z) = I_K/K$, the conditional law is exactly the iid Gaussian design of the paper. $\square$

6 Decoder as a Learned Preconditioner

Now set the decoder attention to be linear, without softmax. Given an encoded linear system $(\widehat{G}, \widehat{b})$, define the empirical cross-correlation

$$s := \frac{K}{M} \widehat{G}^\top \widehat{b}.$$

For a query vector $q \in \mathbb{R}^K$, the paper's linear-attention feature is

$$H_q = qs^\top.$$

The decoder prediction is

$$\widehat{y}(q) = \text{tr}(W_{\text{dec}} H_q^\top) = q^\top W_{\text{dec}} s.$$

Thus the decoder's recovered latent vector is

$$\widehat{z} = W_{\text{dec}} s = W_{\text{dec}} \left(\frac{K}{M} \widehat{G}^\top \widehat{b} \right).$$

Hence W_{dec} is a learned preconditioner. In the paper’s exact task-independent Gaussian design, the population normal matrix is the identity:

$$\mathbb{E} \left[\frac{K}{M} G^\top G \mid z \right] = I_K.$$

For the GP-forced PDE prompt law, the exact population normal matrix is instead

$$\mathbb{E} \left[\frac{K}{M} G^\top G \mid z \right] = \frac{K}{R} \sum_{r=1}^R H_r(z),$$

so the learned preconditioner is adapted to this task-dependent geometry. In the paper’s constant-design case, the replica calculation gives

$$W_{\text{dec}}^\star \stackrel{d}{=} \left(\sqrt{\widehat{\chi}} T + \sqrt{\widehat{\chi}} R + \widehat{m} S \right) (\widehat{q} S + (\lambda + \widehat{q}) I_K)^{-1},$$

where

$$S = \frac{1}{M_0} \sum_{\mu=1}^{M_0} w^\mu (w^\mu)^\top$$

is the task covariance and the hatted parameters are the conjugate order parameters of the paper. This formula shows explicitly that linear attention learns a spectral preconditioner adapted to the low-rank task covariance S .

Assumption 1 (bounded learned preconditioner). *On the high-probability replica event,*

$$\|W_{\text{dec}}^\star\|_{\text{op}} \leq B_{\text{dec}}.$$

Proposition 3 (decoder latent bound). *Assume the decoder is linear attention and the encoded rows are sampled from the exact GP prompt law of Proposition 1. Let*

$$e := \widehat{b} - \widehat{G}z$$

be the encoder residual. Define the exact GP row-transfer constant

$$C_{\text{GP}} := \frac{\mathbb{E} \left[\frac{K}{M^2} \|\widehat{G}^\top e\|_2^2 \right]}{\mathbb{E} \left[\frac{1}{M} \|e\|_2^2 \right]},$$

with the convention $C_{\text{GP}} = 0$ if the denominator is zero. Under the replica saddle-point prediction for the exact GP prompt law and Assumption 1,

$$\boxed{\frac{1}{K} \mathbb{E} \|\widehat{z} - z\|_2^2 \leq E_{\Pi, \text{dec}}^{\text{rep}, \text{GP}} + B_{\text{dec}}^2 C_{\text{GP}} \mathcal{E}_{\text{enc}}.}$$

Combining with Proposition 2,

$$\boxed{\frac{1}{K} \mathbb{E} \|\widehat{z} - z\|_2^2 \leq E_{\Pi, \text{dec}}^{\text{rep}, \text{GP}} + B_{\text{dec}}^2 C_{\text{GP}} E_{\Pi, \text{enc}}^{\text{rep}, \text{GP}}.}$$

If the design is exactly isotropic iid with row covariance I_K/K and the residual is conditionally independent of the rows, then $C_{\text{GP}} = K/M = 1/\widetilde{\alpha}_{\text{dec}}$, recovering the paper’s prompt scaling.

Proof. Write the decoder statistic as

$$s = \frac{K}{M} \widehat{G}^\top \widehat{b} = \frac{K}{M} \widehat{G}^\top \widehat{G}z + \frac{K}{M} \widehat{G}^\top e.$$

The first term is the noiseless prompt statistic analyzed by the exact GP replica calculation. Therefore the scalar prediction error $\mathbb{E}_q[(q^\top W_{\text{dec}} s_0 - q^\top z)^2]$ equals $E_{\Pi, \text{dec}}^{\text{rep, GP}}$, where $s_0 = (K/M)\widehat{G}^\top \widehat{G}z$. Since $q \sim N(0, I_K/K)$,

$$\mathbb{E}_q[(q^\top a)^2] = \frac{1}{K} \|a\|_2^2,$$

so the noiseless latent contribution is $E_{\Pi, \text{dec}}^{\text{rep, GP}}$ after normalization by K .

For the residual term, no isotropic approximation is used:

$$\frac{1}{K} \mathbb{E} \left\| W_{\text{dec}} \frac{K}{M} \widehat{G}^\top e \right\|_2^2 \leq B_{\text{dec}}^2 \mathbb{E} \left[\frac{K}{M^2} \|\widehat{G}^\top e\|_2^2 \right] = B_{\text{dec}}^2 C_{\text{GP}} \mathcal{E}_{\text{enc}}.$$

The two contributions give the claim. $\square$

7 PDE Solution Bound

Let the final reconstructed operator be

$$\widehat{A}(\widehat{z}) = \widehat{A}_0 + \sum_{k=1}^K \widehat{z}_k \widehat{A}_k.$$

Because the dictionary is identifiable only up to a GL_K gauge, assume we have fixed the best-conditioned gauge and write

$$\epsilon_{\text{dict}} := \left\| \widehat{A}_0 - A_0 \right\|_{\text{op}} + \|z\|_2 \left(\sum_{k=1}^K \|\widehat{A}_k - A_k\|_{\text{op}}^2 \right)^{1/2},$$

and

$$B_\Psi^2 := \sum_{k=1}^K \|\widehat{A}_k\|_{\text{op}}^2.$$

Then

$$\|\widehat{A}(\widehat{z}) - A(z)\|_{\text{op}} \leq \epsilon_{\text{dict}} + B_\Psi \|\widehat{z} - z\|_2.$$

Assumption 2 (ellipticity). *For all admissible z , $A(z)$ is symmetric positive definite and*

$$\lambda_{\min}(A(z)) \geq \mu > 0.$$

On the event considered,

$$\|\widehat{A}(\widehat{z}) - A(z)\|_{\text{op}} \leq \mu/2.$$

Theorem 1 (end-to-end bound). *Under Proposition 1, Assumptions 1 and 2, and the replica saddle-point prediction for the exact GP prompt law,*

$$\mathbb{E} \frac{\|\widehat{u}_\star - u_\star\|_2^2}{\|u_\star\|_2^2} \leq \frac{8}{\mu^2} \mathbb{E} \epsilon_{\text{dict}}^2 + \frac{8B_\Psi^2 K}{\mu^2} \left(E_{\Pi, \text{dec}}^{\text{rep, GP}} + B_{\text{dec}}^2 C_{\text{GP}} E_{\Pi, \text{enc}}^{\text{rep, GP}} \right).$$

If the final solve uses ridge $S_\gamma(B)f = (B^\top B + \gamma I)^{-1} B^\top f$, add the usual deterministic bias term

$$2 \frac{\gamma^2}{(\mu^2 + \gamma)^2}$$

and replace $2/\mu$ in the proof by the Lipschitz constant of S_γ on the uniformly elliptic operator class.

Proof. For the exact solve,

$$\hat{u}_\star - u_\star = \hat{A}(\hat{z})^{-1}(A(z) - \hat{A}(\hat{z}))u_\star.$$

Since $\lambda_{\min}(\hat{A}(\hat{z})) \geq \mu/2$,

$$\frac{\|\hat{u}_\star - u_\star\|_2}{\|u_\star\|_2} \leq \frac{2}{\mu} \|\hat{A}(\hat{z}) - A(z)\|_{\text{op}}.$$

Using

$$\|\hat{A}(\hat{z}) - A(z)\|_{\text{op}}^2 \leq 2\epsilon_{\text{dict}}^2 + 2B_\Psi^2 \|\hat{z} - z\|_2^2$$

and Proposition 3 proves the theorem. $\square$

8 Training Dynamics of the Encoder and Decoder

To fit exactly into the linear-attention training-dynamics framework, remove the auxiliary losses L_u, L_a, L_A during this analysis. Train the encoder with the prompt consistency loss and train the decoder with the latent least-squares loss:

$$L_{\text{enc}} = \mathbb{E} \left[\frac{1}{M} \|\hat{G}_\theta z - \hat{b}_\theta\|_2^2 \right], \quad L_{\text{dec}} = \frac{1}{K} \mathbb{E} \|\hat{z}_\theta - z\|_2^2.$$

The extra physical losses are useful experimentally, but they obscure the exact correspondence with linear regression ICL.

8.1 Collapsed Effective Dynamics

Let $P_{\text{enc}}(t)$ and $P_{\text{dec}}(t)$ denote the effective linear maps implemented by the trained linear-attention blocks after collapsing the value and key-query factors. This is the analogue of the effective matrix in the linear-attention dynamics papers.

For a context matrix G_c with labels $b_c = G_c z$, define the context correlation statistic

$$s_c := \frac{K}{M_c} G_c^\top b_c = \frac{K}{M_c} G_c^\top G_c z.$$

For an encoder query weak row q , the linear-attention encoder predicts

$$\hat{b}(q, t) = q^\top P_{\text{enc}}(t) s_c.$$

The exact encoder training loss is therefore

$$L_{\text{enc}}(P) = \mathbb{E} \left[(q^\top z - q^\top P s_c)^2 \right].$$

Let

$$H_q(z) := \mathbb{E}[qq^\top | z], \quad C_{zs}^{\text{enc}} := \mathbb{E}[H_q(z) z s_c^\top], \quad C_{ss}^{\text{enc}}(P) := \mathbb{E}[H_q(z) P s_c s_c^\top].$$

Then the exact gradient-flow dynamics of the collapsed encoder map is

$$\boxed{\tau_{\text{enc}} \dot{P}_{\text{enc}} = C_{zs}^{\text{enc}} - C_{ss}^{\text{enc}}(P_{\text{enc}}).}$$

If the query row covariance is task-independent, $H_q(z) = H_q$, this becomes the closed linear matrix ODE

$$\tau_{\text{enc}} \dot{P}_{\text{enc}} = H_q C_{zs} - H_q P_{\text{enc}} C_{ss}, \quad C_{zs} := \mathbb{E}[z s_c^\top], \quad C_{ss} := \mathbb{E}[s_c s_c^\top].$$

When H_q is invertible, the fixed point is

$$P_{\text{enc}}^\star = C_{zs} C_{ss}^{-1}.$$

Thus encoder training learns the preconditioner that maps the context correlation statistic s_c back to the task z ; consistency $\widehat{G}z - \widehat{b} \approx 0$ is obtained because the predicted row label is $q^\top P_{\text{enc}}^* s_c \approx q^\top z$.

For the decoder, the statistic is

$$s := \frac{K}{M} \widehat{G}^\top \widehat{b}.$$

The decoder always has the least-squares form

$$\widehat{z}(t) = P_{\text{dec}}(t)s.$$

Training changes only the preconditioner $P_{\text{dec}}(t)$. The exact collapsed decoder loss is

$$L_{\text{dec}}(P) = \frac{1}{K} \mathbb{E} \|z - Ps\|_2^2,$$

and gradient flow gives

$$\tau_{\text{dec}} \dot{P}_{\text{dec}} = C_{zs}^{\text{dec}} - P_{\text{dec}} C_{ss}^{\text{dec}}, \quad C_{zs}^{\text{dec}} := \mathbb{E}[zs^\top], \quad C_{ss}^{\text{dec}} := \mathbb{E}[ss^\top].$$

Hence

$$P_{\text{dec}}^* = C_{zs}^{\text{dec}} (C_{ss}^{\text{dec}})^{-1},$$

and

$$P_{\text{dec}}(t) = P_{\text{dec}}^* + (P_{\text{dec}}(0) - P_{\text{dec}}^*) \exp(-C_{ss}^{\text{dec}} t / \tau_{\text{dec}})$$

whenever C_{ss}^{dec} is constant and diagonalizable. In an eigenbasis of C_{ss}^{dec} , each mode obeys

$$p_a(t) = p_a^* + (p_a(0) - p_a^*) e^{-\lambda_a t / \tau_{\text{dec}}}.$$

Large-eigenvalue directions of the prompt statistic are learned first; weakly observed GP/PDE directions are learned slowly or not at all.

8.2 Factorized Linear-Attention Dynamics

The collapsed ODE above describes the algorithm learned by the effective map. The actual linear-attention parameters are factorized. This factorization is what creates the abrupt or staged dynamics observed in the training-dynamics papers [2, 3].

For merged key-query linear attention, Zhang et al. [2] show that the model is equivalent to a two-layer linear network on a cubic feature map and that training has two fixed points: the zero map and the global ICL solution. From small initialization, the loss has a long plateau followed by one abrupt drop. In our notation, that abrupt drop is the moment when $P_{\text{enc}}(t)$ or $P_{\text{dec}}(t)$ rapidly moves from near zero to the preconditioning map $C_{zs} C_{ss}^{-1}$.

For separate key and query, the same paper shows exponentially many fixed points and saddle-to-saddle dynamics. The model learns principal components sequentially. Translating to the PDE decoder, during the $(m+1)$ -st plateau,

$$\widehat{z}(t) \approx \sum_{a=1}^m p_a^* e_a e_a^\top s,$$

where e_a are eigenvectors of the prompt-statistic covariance. This is principal-component least squares: the decoder is still applying a least-squares-style normal-equation statistic, but its preconditioner has rank m . After all active modes are learned, the decoder becomes the full learned least-squares preconditioner.

This staged spectral picture matches exact linear-transformer dynamics: timescale separation is controlled by the data covariance [3]. It also matches softmax-attention analyses: anisotropic covariates lead to learned preconditioned gradient descent [4].

For GP-forced PDE prompts, the ordering of the stages is governed by the spectrum of the exact observable normal statistic

$$\Gamma(z) := \frac{K}{M} G_z^\top G_z, \quad \bar{\Gamma} := \mathbb{E}[\Gamma(z)].$$

Smooth GP kernels suppress high-frequency forcing modes, which shrinks the corresponding eigenvalues of $\bar{\Gamma}$ and delays those coefficient directions in training. This is the exact sense in which the GP kernel changes training difficulty: it changes the spectrum of the statistics that drive the preconditioner ODE.

8.3 Training-Dynamics Summary

The encoder and decoder have the same dynamical skeleton:

$$\text{context correlations } s \longrightarrow \text{learned preconditioner } P(t) \longrightarrow \text{task estimate } P(t)s.$$

The encoder uses this skeleton to make the prompt system consistent, $\hat{G}z - \hat{b} \rightarrow 0$. The decoder uses the same skeleton to solve the normal equations. At every time, the decoder has the form

$$\hat{z}(t) = P_{\text{dec}}(t) \frac{K}{M} \hat{G}^\top \hat{b}.$$

So the algorithmic primitive is always least squares; training dynamics only determines which spectral preconditioner has been learned, and how many observable task directions have entered that preconditioner.

9 Exact MP Training Curves in the Isotropic Design Limit

This section gives the fully explicit training curves in the regime where the weak-row design is exactly isotropic Gaussian on the active task subspace. Let $B \in \mathbb{R}^{K \times r}$ have orthonormal columns, $r = r_z$, and write $z = B\zeta$ with

$$\zeta \sim N\left(0, \frac{K}{r} I_r\right).$$

Assume

$$GB = \frac{1}{\sqrt{K}} X, \quad X_{ja} \stackrel{\text{iid}}{\sim} N(0, 1), \quad M = mR.$$

The normalized active Gram matrix is

$$\Gamma := \frac{K}{M} B^\top G^\top GB = \frac{1}{M} X^\top X \in \mathbb{R}^{r \times r}.$$

The decoder statistic restricted to the active subspace is exactly

$$s_B := B^\top \frac{K}{M} G^\top b = \Gamma \zeta.$$

Theorem 2 (MP spectrum and exact preconditioner curve). *Let*

$$\tilde{\alpha} := \frac{M}{r}, \quad c := \frac{r}{M} = \tilde{\alpha}^{-1}.$$

As $M, r \rightarrow \infty$ with $r/M \rightarrow c$, the empirical spectral distribution of Γ converges almost surely to the Marchenko–Pastur law

$$\mu_{\text{MP},c} = (1 - \tilde{\alpha})_+ \delta_0 + \frac{\tilde{\alpha}}{2\pi\lambda} \sqrt{(\lambda_+ - \lambda)(\lambda - \lambda_-)} \mathbf{1}_{[\lambda_-, \lambda_+]}(\lambda) d\lambda,$$

where

$$\lambda_{\pm} = (1 \pm \sqrt{c})^2 = \left(1 \pm \frac{1}{\sqrt{\tilde{\alpha}}}\right)^2.$$

Equivalently, the nonzero squared singular values of the raw active design GB are $(M/K)\lambda$ with $\lambda \sim \mu_{\text{MP},c}$. Train the collapsed decoder preconditioner with

$$L(P) = \frac{1}{K} \mathbb{E}_{\zeta} \|\zeta - P\Gamma\zeta\|_2^2 = \frac{1}{r} \text{tr}(I - P\Gamma)(I - P\Gamma)^{\top}$$

and gradient flow

$$\tau \dot{P} = \Gamma - P\Gamma^2, \quad P(0) = 0.$$

Then, for every finite r, M ,

$$P(t) = \Gamma^{\dagger}(I - e^{-\Gamma^2 t/\tau}), \quad P(t)\Gamma = \Pi_{\text{range } \Gamma}(I - e^{-\Gamma^2 t/\tau}).$$

Consequently the exact finite-dimensional training curve is

$$\mathcal{E}_r(t) := \frac{1}{K} \mathbb{E}_{\zeta} \|\zeta - P(t)\Gamma\zeta\|_2^2 = \frac{1}{r} \text{tr} e^{-2\Gamma^2 t/\tau}.$$

In the MP limit,

$$\boxed{\mathcal{E}(t) = \int e^{-2\lambda^2 t/\tau} \mu_{\text{MP},c}(d\lambda).}$$

Proof. Since $GB = K^{-1/2}X$,

$$\Gamma = \frac{K}{M} B^{\top} G^{\top} GB = \frac{1}{M} X^{\top} X,$$

so the Marchenko–Pastur theorem gives the stated limit. The loss identity follows from $\mathbb{E}[\zeta\zeta^{\top}] = (K/r)I_r$:

$$\frac{1}{K} \mathbb{E} \|\zeta - P\Gamma\zeta\|_2^2 = \frac{1}{r} \text{tr}(I - P\Gamma)(I - P\Gamma)^{\top}.$$

Differentiating this quadratic loss gives

$$\tau \dot{P} = \Gamma - P\Gamma^2.$$

Diagonalize $\Gamma = U \text{diag}(\lambda_a) U^{\top}$. Since $P(0) = 0$, the solution remains diagonal in this eigenbasis. Each eigenmode satisfies

$$\tau \dot{p}_a = \lambda_a - p_a \lambda_a^2, \quad p_a(0) = 0.$$

Thus, if $\lambda_a > 0$,

$$p_a(t) = \lambda_a^{-1}(1 - e^{-\lambda_a^2 t/\tau}), \quad p_a(t)\lambda_a = 1 - e^{-\lambda_a^2 t/\tau},$$

while if $\lambda_a = 0$, $p_a(t)\lambda_a = 0$. Therefore the residual eigenvalue is $1 - p_a(t)\lambda_a = e^{-\lambda_a^2 t/\tau}$, including zero modes. Averaging the squared residual eigenvalues gives the finite curve; the MP limit follows by weak convergence of the empirical spectral measure. $\square$

Corollary 1 (spectrum of the learned normal-equation map). *The empirical spectrum of $P(t)\Gamma$ is the pushforward of $\mu_{\text{MP},c}$ by*

$$\phi_t(\lambda) = 1 - e^{-\lambda^2 t/\tau}.$$

Equivalently, away from atoms, with

$$\lambda(y) = \sqrt{-\frac{\tau}{t} \log(1 - y)}, \quad y \in \phi_t([\lambda_-, \lambda_+]),$$

the density is

$$\rho_{P\Gamma,t}(y) = \rho_{\text{MP},c}(\lambda(y)) \frac{\tau}{2t \lambda(y)(1 - y)}.$$

As $t \rightarrow \infty$, $P(t)\Gamma$ converges to the projector onto $\text{range } \Gamma$. Hence the limiting error is

$$\lim_{t \rightarrow \infty} \mathcal{E}(t) = (1 - \tilde{\alpha})_+.$$

Corollary 2 (early and late time laws). *At small time,*

$$\mathcal{E}(t) = 1 - \frac{2t}{\tau} \int \lambda^2 \mu_{\text{MP},c}(d\lambda) + O(t^2) = 1 - \frac{2t}{\tau}(1 + c) + O(t^2).$$

If $\tilde{\alpha} > 1$, then $\lambda_- > 0$ and

$$\mathcal{E}(t) \asymp \exp(-2\lambda_-^2 t/\tau) t^{-3/2} \quad (t \rightarrow \infty).$$

If $\tilde{\alpha} = 1$, then $\lambda_- = 0$ and

$$\mathcal{E}(t) \asymp t^{-1/4}.$$

If $\tilde{\alpha} < 1$, the curve converges to the unobservable-dimensional plateau $1 - \tilde{\alpha}$.

Remark 2. *The same theorem applies to the encoder after replacing M by the number of weak rows in the encoder context and τ by τ_{enc} . Its curve is the consistency curve for $\hat{G}z - \hat{b}$. For the exact GP-forced PDE law, the MP law is replaced by the spectral law of the deformed sample covariance with population covariance $H(z)$. Thus the formulas above are exact for the isotropic reduction and become deformed-MP formulas for the full GP geometry.*

10 What Has Been Reused

The imported statistical-mechanics step is the one from [1]: the trained linear-attention matrix is replaced by a replica quadratic surrogate, whose solution is a spectral shrinkage or preconditioning map. In the exact GP-forcing PDE model, the surrogate is averaged over the task-conditional prompt law P_z , not over a task-independent iid Gaussian design. The paper’s closed forms are recovered when $H_r(z) = I_K/K$ exactly.

Everything specific to the PDE prompt law is exact:

$$f \sim GP \Rightarrow u = A(z)^{-1}f \text{ Gaussian conditional on } z \Rightarrow g = (\langle A_k u, v \rangle)_{k=1}^K \text{ Gaussian} \Rightarrow b = g^\top z.$$

Thus the encoder residual is a full-prompt low-rank ICL prediction error for the task z , and the decoder is a learned-preconditioner problem applied to the normal-equation statistic $(K/M)G^\top b$.

Remark 3 (practical reading). *The theorem predicts that improving the encoder residual helps the decoder only through the exact factor C_{GP} . In the isotropic iid design case, $C_{\text{GP}} = K/M$; for GP-forced PDE prompts, C_{GP} is determined by the actual weak-row geometry. The hard task-diversity transition is still governed by whether the number of distinct pretraining tasks M_0 is large enough relative to the latent rank r_z .*

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